

## UNIT 3: ACCOUNTING STANDARD 17 SEGMENT REPORTING

### LEARNING OUTCOMES

**After studying this unit, you will be able to comprehend the-**

- ♦ Definition and Identification of Reportable Segments
- ♦ Primary and Secondary Segment Reporting Formats
- ♦ Business and Geographical Segments
- ♦ How to identify the Reportable Segments
- ♦ Disclosures.



### 3.1 INTRODUCTION

AS 17 is mandatory in respect of non-SMCs (and level I entities in case of non-corporates). Other entities are encouraged to comply with AS 17.

This standard establishes principles for reporting financial information about different types of products and services an enterprise produces and different geographical areas in which it operates. The standard is more relevant for assessing risks and returns of a diversified or multi-locational enterprise which may not be determinable from the aggregated data.

Before we start the standard, let us lay down the areas to be covered from the examination point of view.

**Identify the Segments - Business or Geographical**

**Identify the Reportable Segments**

**Prepare a Segmental Report + Make appropriate Disclosures**

### 3.2 OBJECTIVE

Many enterprises provide groups of products and services or operate in geographical areas that are subject to differing rates of profitability, opportunities for growth, future prospects, and risks. The objective of this Standard is to establish principles for reporting financial information, about the different types of products and services an enterprise produces and the different geographical areas in which it operates. Such information helps users of financial statements:

- (a) Better understand the performance of the enterprise;
- (b) Better assess the risks and returns of the enterprise; and
- (c) Make more informed judgements about the enterprise as a whole.

### 3.3 SCOPE

AS 17 should be applied in presenting general purpose financial statements.

An enterprise should comply with the requirements of this Standard fully and not selectively. If a single financial report contains both consolidated financial statements and separate financial statements of the parent, segment information need be presented only on the basis of the consolidated financial statements.



### 3.4 DEFINITION OF THE TERMS USED IN THE ACCOUNTING STANDARD

**A business segment** is a distinguishable component of an enterprise that is engaged in providing an individual product or service or a group of related products or services and that is subject to risks and returns that are different from those of other business segments. Factors that should be considered in determining whether products or services are related include:

- (a) The nature of the products or services
- (b) The nature of the production processes
- (c) The type or class of customers for the products or services
- (d) The methods used to distribute the products or provide the services
- (e) If applicable, the nature of the regulatory environment, for example, banking, insurance, or public utilities

A single business segment does not include products and services with significantly differing risks and returns. While there may be dissimilarities with respect to one or several of the factors listed in the definition of business segment, the products and services included in a single business segment are expected to be similar with respect to a majority of the factors.

**A geographical segment** is a distinguishable component of an enterprise that is engaged in providing products or services within a particular economic environment and that is subject to risks and returns that are different from those of components operating in other economic environments. Factors that should be considered in identifying geographical segments include:

- (a) Similarity of economic and political conditions.
- (b) Relationships between operations in different geographical areas.
- (c) Proximity of operations.
- (d) Special risks associated with operations in a particular area.
- (e) Exchange control regulations and
- (f) The underlying currency risks.

A single geographical segment does not include operations in economic environments with significantly differing risks and returns. A geographical segment may be a single country, a group of two or more countries, or a region within a country.

The risks and returns of an enterprise are influenced both by the geographical location of its operations (where its products are produced or where its service rendering activities are based) and also by the location of its customers (where its products are sold or services are rendered). The definition allows geographical segments to be based on either:

- (a) The location of production or service facilities and other assets of an enterprise; or
- (b) The location of its customers.

The predominant sources of risks affect how most enterprises are organised and managed. Therefore, the organisational structure of an enterprise and its internal financial reporting system are normally the basis for identifying its segments.

**A reportable segment** is a business segment or a geographical segment identified on the basis of foregoing definitions for which segment information is required to be disclosed by AS 17.

**Segment revenue** is the aggregate of

- (i) The portion of enterprise revenue that is directly attributable to a segment;
- (ii) The relevant portion of enterprise revenue that can be allocated on a reasonable basis to a segment; and
- (iii) Revenue from transactions with other segments of the enterprise.

**Segment revenue does not include:**

- (a) Extraordinary items as defined in AS 5;
- (b) Interest or dividend income, including interest earned on advances or loans to other segments unless the operations of the segment are primarily of a financial nature; and
- (c) Gains on sales of investments or on extinguishment of debt unless the operations of the segment are primarily of a financial nature.

**Segment expense** is the aggregate of

- (i) The expense resulting from the operating activities of a segment that is directly attributable to the segment;
- (ii) The relevant portion of enterprise expense that can be allocated on a reasonable basis to the segment; and
- (iii) Including expense relating to transactions with other segments of the enterprise.

**Segment expense does not include:**

- (a) Extraordinary items as defined in AS 5;
- (b) Interest expense, including interest incurred on advances or loans from other segments, unless the operations of the segment are primarily of a financial nature;
- (c) Losses on sales of investments or losses on extinguishment of debt unless the operations of the segment are primarily of a financial nature;
- (d) Income tax expense; and
- (e) General administrative expenses, head-office expenses, and other expenses that arise at the enterprise level and relate to the enterprise as a whole. However, costs are sometimes incurred at the enterprise level on behalf of a segment. Such costs are part of segment expense if they relate to the operating activities of the segment and if they can be directly attributed or allocated to the segment on a reasonable basis.

**Segment result** is segment revenue less segment expense.

**Segment assets** are those operating assets that are employed by a segment in its operating activities and that either are directly attributable to the segment or can be allocated to the segment on a reasonable basis.

If the segment result of a segment includes interest or dividend income, its segment assets include the related receivables, loans, investments, or other interest or dividend generating assets.

Segment assets do not include:

- ♦ income tax assets; and
- ♦ assets used for general enterprise or head-office purposes.

Segment assets are determined after deducting related allowances/provisions that are reported as direct offsets in the balance sheet of the enterprise.

**Segment liabilities** are those operating liabilities that result from the operating activities of a segment and that either are directly attributable to the segment or can be allocated to the segment on a reasonable basis.

If the segment result of a segment includes interest expense, its segment liabilities include the related interest-bearing liabilities.

Examples of segment liabilities include trade and other payables, accrued liabilities, customer advances, product warranty provisions, and other claims relating to the provision of goods and services.

Segment liabilities do not include:

- ♦ income tax liabilities; and
- ♦ borrowings and other liabilities that are incurred for financing rather than operating purposes.

Assets and liabilities that relate jointly to two or more segment should be allocated to segments if, and only if, their related revenues and expenses also are allocated to those segments.

### 3.5 TREATMENT OF INTEREST FOR DETERMINING SEGMENT EXPENSE

The interest expense relating to overdrafts and other operating liabilities identified to a particular segment should not be included as a part of the segment expense unless the operations of the segment are primarily of a financial nature or unless the interest is included as a part of the cost of inventories.

In case interest is included as a part of the cost of inventories where it is so required as per AS 16, read with AS 2 (Revised), and those inventories are part of segment assets of a particular segment, such interest should be considered as a segment expense. In this case, the amount of such interest and the fact that the segment result has been arrived at after considering such interest should be disclosed by way of a note to the segment result.

### 3.6 ALLOCATION

An enterprise looks to its internal financial reporting system as the starting point for identifying those items that can be directly attributed, or reasonably allocated, to segments. There is thus a presumption that amounts that have been identified with segments for internal financial reporting purposes are directly attributable or reasonably allocable to segments for the purpose of measuring the segment revenue, segment expense, segment assets, and segment liabilities of reportable segments.

In some cases, however, a revenue, expense, asset or liability may have been allocated to segments for internal financial reporting purposes on a basis that is understood by enterprise management but that could be deemed arbitrary in the perception of external users of financial statements. Conversely, an enterprise may choose not to allocate some item of revenue, expense, asset or liability for internal financial reporting purposes, even though a reasonable basis for doing so exists. Such an item is allocated pursuant to the definitions of segment revenue, segment expense, segment assets, and segment liabilities in AS 17.

Segment revenue, segment expense, segment assets and segment liabilities are determined before intra-enterprise balances and intra-enterprise transactions are eliminated as part of the process of preparation of enterprise financial statements, except to the extent that such intra-enterprise balances and transactions are within a single segment.

While the accounting policies used in preparing and presenting the financial statements of the enterprise as a whole are also the fundamental segment accounting policies, segment accounting policies include, in addition, policies that relate specifically to segment reporting, such as identification of segments, method of pricing inter-segment transfers, and basis for allocating revenues and expenses to segments.

### 3.7 PRIMARY AND SECONDARY SEGMENT REPORTING FORMATS

The dominant source and nature of risks and returns of an enterprise should govern whether its primary segment reporting format will be business segments or geographical segments.

If the risks and returns of an enterprise are affected predominantly by differences in the products and services it produces, its primary format for reporting segment information should be business segments, with secondary information reported geographically. Similarly, if the risks and returns of the enterprise are affected by the fact that it operates in different countries or other geographical areas, its primary format for reporting segment information should be geographical segments, with secondary information reported for groups of related products and services.

### 3.8 BUSINESS AND GEOGRAPHICAL SEGMENTS

Generally, business and geographical segments are determined on the basis of internal financial reporting to the board of directors and the Chief Executive Officer. But if such segment does not satisfy the definitions given in AS, then following points should be considered:

- (a) If one or more of the segments reported internally to the directors and management is a business segment or a geographical segment based on the factors in the definitions but others are not, paragraph below should be applied only to those internal segments that do not meet the definitions (that is, an internally reported segment that meets the definition should not be further segmented).
- (b) For those segments reported internally to the directors and management that do not satisfy the definitions, management of the enterprise should look to the next lower level of internal segmentation that reports information along product and service lines or geographical lines, as appropriate under the definitions and
- (c) If such an internally reported lower-level segment meets the definition of business segment or geographical segment, the criteria for identifying reportable segments should be applied to that segment.



### 3.9 IDENTIFYING REPORTABLE SEGMENTS (QUANTITATIVE THRESHOLDS)

A business segment or geographical segment should be identified as a reportable segment if:

- (a) Its revenue from sales to external customers and from transactions with other segments is 10% or more of the total revenue, external and internal, of all segments; or
- (b) Its segment result, whether profit or loss, is 10% or more of –
  - (i) The combined result of all segments in profit, or
  - (ii) The combined result of all segments in loss,Whichever is greater in absolute amount; or

(c) Its segment assets are 10% or more of the total assets of all segments. A business segment or a geographical segment which is not a reportable segment as per above paragraph, may be designated as a reportable segment despite its size at the discretion of the management of the enterprise. If that segment is not designated as a reportable segment, it should be included as an unallocated reconciling item.

If total external revenue attributable to reportable segments constitutes less than 75% of the total enterprise revenue, additional segments should be identified as reportable segments, even if they do not meet the 10% thresholds, until at least 75% of total enterprise revenue is included in reportable segments.

**We can summarize the steps as under: Step I - Apply 10% Test**

| Any 1 Test needs to be met - For Reportable Segment: |                                                                                                                        |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Revenue Test</b>                                  | Revenue (External + Internal) of the segment is 10% or more of the Total Revenue of all segments                       |
| <b>Profit/Loss Test</b>                              | <b>Case I - All segments have profits:</b><br>Profit of the segment is 10% or more of the total profit of all segments |

|                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                               | <p><b>Case II - Few segments have profit + Few segments have losses:</b></p> <ol style="list-style-type: none"> <li>1. Add the profits of profitable segments only.</li> <li>2. Add the losses of loss-making segments only.</li> <li>3. Take the figure (from 1 and 2) whichever is greater (in absolute values).</li> <li>4. The segment which has profit/loss equal to 10% or more of the absolute figure computed in Point 3 becomes reportable.</li> </ol> |
| <b>Asset Test</b>                                                                                                                                                                                                                                             | The segment assets are 10% or more of the total assets of all segments.                                                                                                                                                                                                                                                                                                                                                                                         |
| <p><b>Note:</b></p> <p>A business segment or a geographical segment which is not a reportable segment as per above steps, <b>may be designated</b> as a reportable segment despite its size at the <b>discretion of the management</b> of the enterprise.</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

### **Step II - Apply 75% Test (Overall Test):**

Ensure that the **total external revenue** attributable to reportable segments constitutes at least 75% of the **total enterprise revenue**.

If not, additional segments should be identified as reportable segments, even if they do not meet the 10% thresholds, until at least 75% of total enterprise revenue is included in reportable segments.

### **Notes:**

1. A segment identified as a reportable segment in the immediately preceding period because it satisfied the relevant 10% thresholds should continue to be a reportable segment for the current period notwithstanding that its revenue, result, and assets all no longer meet the 10% thresholds.
2. If a segment is identified as a reportable segment in the current period because it satisfies the relevant 10% thresholds, preceding-period segment data that is presented for comparative purposes should, unless it is impracticable to do so, be restated to reflect the newly reportable segment as a separate segment, even if that segment did not satisfy the 10%

thresholds in the preceding period.

### 3.10 SEGMENT ACCOUNTING POLICIES

Segment information should be prepared in conformity with the accounting policies adopted for preparing and presenting the financial statements of the enterprise as a whole. AS 17 does not prohibit the disclosure of additional segment information that is prepared on a basis other than the accounting policies adopted for the enterprise financial statements provided that-

- (a) the information is reported internally to the board of directors and the chief executive officer for purposes of making decisions about allocating resources to the segment and assessing its performance; and
- (b) the basis of measurement for this additional information is clearly described.

### 3.11 PRIMARY REPORTING FORMAT

An enterprise should disclose the following for each reportable segment:

- (a) Segment revenue, classified into segment revenue from sales to external customers and segment revenue from transactions with other segments;
- (b) Segment result;
- (c) Total carrying amount of segment assets;
- (d) Total amount of segment liabilities;
- (e) Total cost incurred during the period to acquire segment assets that are expected to be used during more than one period (tangible and intangible fixed assets);
- (f) Total amount of expense included in the segment result for depreciation and amortisation in respect of segment assets for the period; and
- (g) Total amount of significant non-cash expenses, other than depreciation and amortisation in respect of segment assets that were included in segment expense and, therefore, deducted in measuring segment result.

An enterprise that reports the amount of cash flows arising from operating, investing and financing activities of a segment need not disclose depreciation and amortisation expense and non-cash expenses.

An enterprise should present a reconciliation between the information disclosed for reportable segments and the aggregated information in the enterprise financial statements.

In presenting the reconciliation:,

- segment revenue should be reconciled to enterprise revenue;
- segment result should be reconciled to enterprise net profit or loss;
- segment assets should be reconciled to enterprise assets; and
- segment liabilities should be reconciled to enterprise liabilities.

### 3.12 SECONDARY SEGMENT INFORMATION

If primary format of an enterprise for reporting segment information is business segments, it should also report the following information:

- (a) Segment revenue from external customers by geographical area based on the geographical location of its customers, for each geographical segment whose revenue from sales to external customers is 10% or more of enterprise revenue;
- (b) The total carrying amount of segment assets by geographical location of assets, for each geographical segment whose segment assets are 10% or more of the total assets of all geographical segments; and
- (c) The total cost incurred during the period to acquire segment assets that are expected to be used during more than one period (tangible and intangible fixed assets) by geographical location of assets, for each geographical segment whose segment assets are 10% or more of the total assets of all geographical segments.

If primary format of an enterprise for reporting segment information is geographical segments (whether based on location of assets or location of customers), it should also report the following segment information for each business segment whose revenue from sales to external customers is 10% or more

of enterprise revenue or whose segment assets are 10% or more of the total assets of all business segments:

- a. Segment revenue from external customers;
- b. The total carrying amount of segment assets; and
- c. The total cost incurred during the period to acquire segment assets that are expected to be used during more than one period (tangible and intangible fixed assets).

### 3.13 OTHER DISCLOSURES

In measuring and reporting segment revenue from transactions with other segments, inter-segment transfers should be measured on the basis that the enterprise actually used to price those transfers. The basis of pricing inter-segment transfers and any change therein should be disclosed in the financial statements.

Changes in accounting policies adopted for segment reporting that have a material effect on segment information should be disclosed. Such disclosure should include a description of the nature of the change, and the financial effect of the change if it is reasonably determinable.

Some changes in accounting policies may relate specifically to segment reporting. Example could be:

- ♦ changes in identification of segments; and
- ♦ changes in the basis for allocating revenues and expenses to segments.

Such changes can have a significant impact on the segment information reported but will not change aggregate financial information reported for the enterprise. To enable users to understand the impact of such changes, this Standard requires the disclosure of the nature of the change and the financial effects of the change, if reasonably determinable.

An enterprise should indicate the types of products and services included in each reported business segment and indicate the composition of each reported geographical segment, both primary and secondary, if not otherwise disclosed in the financial statements.

### Illustration 1

The Chief Accountant of Sports Ltd. gives the following data regarding its six segments: **in lakhs**

| Particulars     | M   | N     | O  | P  | Q    | R  | Total |
|-----------------|-----|-------|----|----|------|----|-------|
| Segment Assets  | 40  | 80    | 30 | 20 | 20   | 10 | 200   |
| Segment Results | 50  | (190) | 10 | 10 | (10) | 30 | (100) |
| Segment Revenue | 300 | 620   | 80 | 60 | 80   | 60 | 1,200 |

The Chief accountant is of the opinion that segments "M" and "N" alone should be reported. Is he justified in his view? Discuss.

### Solution

As per AS 17 'Segment Reporting', a business segment or geographical segment should be identified as a reportable segment if:

Its revenue from sales to external customers and from other transactions with other segments is 10% or more of the total revenue- external and internal of all segments; or

Its segment result whether profit or loss is 10% or more of:

- ♦ The combined result of all segments in profit; or
- ♦ The combined result of all segments in loss, whichever is greater in absolute amount; or

Its segment assets are 10% or more of the total assets of all segments.

If the total external revenue attributable to reportable segments constitutes less than 75% of total enterprise revenue, additional segments should be identified as reportable segments even if they do not meet the 10% thresholds until atleast 75% of total enterprise revenue is included in reportable segments.

On the basis of turnover criteria segments M and N are reportable segments.

On the basis of the result criteria, segments M, N and R are reportable segments (since their results in absolute amount is 10% or more of 200 lakhs).

On the basis of asset criteria, all segments except R are reportable segments.

Since all the segments are covered in at least one of the above criteria, all segments have to be reported in accordance with Accounting Standard (AS) 17. Hence the opinion of chief accountant is wrong.

### **Illustration 2**

*A Company has an inter-segment transfer pricing policy of charging at cost less 10%. The market prices are generally 25% above cost. Is the policy adopted by the company correct?*

### **Solution**

AS 17 'Segment Reporting' requires that inter-segment transfers should be measured on the basis that the enterprise actually used to price these transfers. The basis of pricing inter-segment transfers and any change therein should be disclosed in the financial statements. Hence the enterprise can have its own policy for pricing inter-segment transfers and hence inter-segment transfers may be based on cost, below cost or market price. However, whichever policy is followed, the same should be disclosed and applied consistently. Therefore, in the given case inter-segment transfer pricing policy adopted by the company is correct if, followed consistently.

### **Illustration 3**

*M/s XYZ Ltd. has three segments namely X, Y, Z. The total Assets of the Company are ₹ 10.00 crores. Segment X has ₹ 2.00 crores, segment Y has ₹ 3.00 crores and segment Z has ₹ 5.00 crores. Deferred tax assets included in the assets of each segments are X- ₹ 0.50 crores, Y— ₹ 0.40 crores and Z— ₹ 0.30 crores. The accountant contends that all the three segments are reportable segments. Comment.*

### **Solution**

According to AS 17 "Segment Reporting", segment assets do not include income tax assets. Therefore, the revised total assets are ₹ 8.8 crores [ $₹ 10 \text{ crores} - (₹ 0.5 + ₹ 0.4 + ₹ 0.3)$ ]. Segment X holds total assets of ₹ 1.5 crores ( $₹ 2 \text{ crores} - ₹ 0.5 \text{ crores}$ ); Segment Y holds ₹ 2.6 crores ( $₹ 3 \text{ crores} - ₹ 0.4 \text{ crores}$ ); and Segment Z holds ₹ 4.7 crores ( $₹ 5 \text{ crores} - ₹ 0.3 \text{ crores}$ ). Thus all the three segments hold more than 10% of the total assets, all segments are reportable segments.

#### Illustration4

Prepare a segmental report for publication in Diversifiers Ltd. from the following details of the company's three divisions and the head office:

|                              | ₹ ('000)      |
|------------------------------|---------------|
| <i>Forging Shop Division</i> |               |
| Sales to Bright Bar Division | 4,575         |
| Other Domestic Sales         | 90            |
| Export Sales                 | <u>6,135</u>  |
|                              | <u>10,800</u> |
| <i>Bright Bar Division</i>   |               |
| Sales to Fitting Division    | 45            |
| Export Sales to Rwanda       | <u>300</u>    |
|                              | <u>345</u>    |
| <i>Fitting Division</i>      |               |
| Export Sales to Maldives     | <u>270</u>    |

| Particulars                  | Head Office<br>₹ ('000) | Forging Shop Division<br>₹ ('000) | Bright Bar Division<br>₹ ('000) | Fitting Division<br>₹ ('000) |
|------------------------------|-------------------------|-----------------------------------|---------------------------------|------------------------------|
| Pre-tax operating result     |                         | 240                               | 30                              | (12)                         |
| Head office cost reallocated |                         | 72                                | 36                              | 36                           |
| Interest costs               |                         | 6                                 | 8                               | 2                            |
| Fixed assets                 | 75                      | 300                               | 60                              | 180                          |
| Net current assets           | 72                      | 180                               | 60                              | 135                          |
| Long-term liabilities        | 57                      | 30                                | 15                              | 180                          |

**Diversifiers  
Ltd. Segmental  
Report**

(` '000)

| Particulars                                              | Divisions       |               |            | Inter<br>Segment<br>Eliminations | Consolidated<br>Total |
|----------------------------------------------------------|-----------------|---------------|------------|----------------------------------|-----------------------|
|                                                          | Forging<br>shop | Bright<br>Bar | Fitting    |                                  |                       |
| Segment Revenue                                          |                 |               |            |                                  |                       |
| Sales:                                                   |                 |               |            |                                  |                       |
| Domestic                                                 | 90              | ~             | ~          | ~                                | 90                    |
| Export                                                   | <u>6,135</u>    | <u>300</u>    | <u>270</u> | ~                                | <u>6,705</u>          |
| External Sales                                           | 6,225           | 300           | 270        | ~                                | 6,795                 |
| Inter-Segment Sales                                      | <u>4,575</u>    | <u>45</u>     | ~          | <u>4,620</u>                     | ~                     |
| Total Revenue                                            | <u>10,800</u>   | <u>345</u>    | <u>270</u> | <u>4,620</u>                     | <u>6,795</u>          |
| Segment Result<br>(Given)                                | 240             | 30            | (12)       |                                  | 258                   |
| Head Office Expenses                                     |                 |               |            |                                  | <u>(144)</u>          |
| Operating Profit                                         |                 |               |            |                                  | 114                   |
| Interest Expense                                         |                 |               |            |                                  | <u>(16)</u>           |
| Profit Before Tax                                        |                 |               |            |                                  | <u>98</u>             |
| Information in<br>Relation to Assets<br>and Liabilities: |                 |               |            |                                  |                       |
| Fixed Assets                                             | 300             | 60            | 180        | ~                                | 540                   |
| Net Current Assets                                       | <u>180</u>      | <u>60</u>     | <u>135</u> | ~                                | <u>375</u>            |
| Segment assets                                           | <u>480</u>      | <u>120</u>    | <u>315</u> | ~                                | <u>915</u>            |
| Unallocated<br>Corporate Assets (75<br>+ 72)             | ~               | ~             | ~          | ~                                | <u>147</u>            |
| Total assets                                             |                 |               |            |                                  | <u>1,062</u>          |
| Segment liabilities                                      | 30              | 15            | 180        | ~                                | 225                   |
| Unallocated<br>corporate liabilities                     |                 |               |            |                                  | <u>57</u>             |
| Total liabilities                                        |                 |               |            |                                  | <u>282</u>            |

### Sales Revenue by Geographical Market

(` '000)

|                | Home Sales | Export Sales (by forging shop division) | Export to Rwanda | Export to Maldives | Consolidated Total |
|----------------|------------|-----------------------------------------|------------------|--------------------|--------------------|
| External sales | 90         | 6,135                                   | 300              | 270                | 6,795              |

#### Illustration 5

*Microtech Ltd. produces batteries for scooters, cars, trucks, and specialised batteries for invertors and UPS. How many segments should it have and why?*

#### Solution

In case of Microtech Ltd., the basic product is the batteries, but the risks and returns of the batteries for automobiles (scooters, cars and trucks) and batteries for invertors and UPS are affected by different set of factors. In case of automobile batteries, the risks and returns are affected by the Government policy, road conditions, quality of automobiles, etc. whereas in case of batteries for invertors and UPS, the risks and returns are affected by power condition, standard of living, etc. Therefore, it can be said that Microtech Ltd. has two business segments viz- 'Automobile batteries' and 'batteries for Invertors and UPS'.

**Reference:** The students are advised to refer the full text of AS 17 "Segment Reporting".

## TEST YOUR KNOWLEDGE

### Multiple Choice Questions

- As per AS 17, reportable segments are those whose total revenue from external sales and inter-segment sales is
  - 10% or more of the total revenue of all segments
  - 10% or more of the total revenue of all external segments
  - 12% or more of the total revenue of all segments
  - 12% or more of the total revenue of all external segments

2. Which of the following statements is correct?
  - (a) Management has a discretion to include a segment as a reportable segment even if it passes the 10% materiality test.
  - (b) Management has a discretion to include any segment as a reportable segment if it fails the 12% materiality test.
  - (c) It is mandatory for the management to include the segment as a reportable segment if it passes the 10% materiality test.
  - (d) It is not mandatory for the management to include the segment as a reportable segment if it passes the 10% materiality test.
3. Which of the following statements is correct?
  - (a) The overall test of 75% considers only external revenue to compute the threshold limit.
  - (b) The overall test of 75% considers only internal revenue to compute the threshold limit.
  - (c) The overall test of 75% considers both internal and external revenue to compute the threshold limit.
  - (d) It is management choice whether they want to include both external and internal revenue for computing threshold limit.
4. Which of the following statements is correct?
  - (a) The 10% test computed on the basis of revenue, considers both internal and external revenue to compute the threshold limit.
  - (b) The 10% test computed on the basis of revenue, considers only external revenue to compute the threshold limit.
  - (c) The 10% test computed on the basis of revenue, considers only internal revenue to compute the threshold limit.
  - (d) It is management choice whether they want to include both external and internal revenue for computing threshold limit.
5. Which of the following statements is correct?
  - (a) In case of 10% test based on profit/loss, we need to consider that any segment whose profit or loss is 10% or more than the net profit or net

*loss respectively of all segments taken together becomes reportable segment.*

- (b) In case of 10% test based on profit/loss, we need to consider that any segment whose profit or loss is 10% or more than the net profit (after netting the losses) of all segments taken together becomes reportable segment.*
- (c) In case of 10% test based on profit/loss, we need to consider that any segment whose profit or loss is 10% or more than the net profit or loss (whichever is higher in absolute figures) of all segments taken together becomes reportable segment.*
- (d) In case of 10% test based on profit/loss, we need to consider that any segment whose profit or loss is 10% or more than the net profit or loss (whichever is lower in absolute figures) of all segments taken together becomes reportable segment.*

### **Scenario based Question**

- 6. Nathan Limited has three segments namely P, Q and R. The assets of the company are ` 15 crores. Segment P has 4 crores, Segment Q has 6 crores and Segment R has 5 crores. Deferred tax assets included in the assets of each segment are P - ` 1 crore, Q - ` 0.90 crores and R - ` 0.80 crores. The accountant contends all these three segments are reportable segments. Comment.*
- 7. Company A is engaged in the manufacture and sale of products, which constitute two distinct business segments. The products of the Company are sold in the domestic market only. The management information system of the Company is organized to reflect operating information by two broad market segments, rural and urban.*

*Besides the two business segments, how should Company A identify geographical segments? Do geographical segments exist within the same country? Explain in line with the provisions of AS 17.*
- 8. PK Ltd. has identified business segment as its primary reporting format. It has identified India, USA and UK as three geographical segments. It sells its products in the Indian market, which constitutes 70 percent of the Company's*

sales. 25 per cent is sold in USA and the balance is sold in UK.

Is PK Ltd. as part of its geographical secondary segment information, required to disclose segment revenue from export sales, where such sales are not significant?

9. XYZ Ltd. has 5 business segments. Profit / Loss of each of the segments for the year ended 31st March, 20X2 have been provided below. You are required to identify from the following whether reportable segments or not reportable segments, on the basis of "profitability test" as per AS-17.

| Segment | Profit (Loss) ` in lakhs |
|---------|--------------------------|
| A       | 225                      |
| B       | 25                       |
| C       | (175)                    |
| D       | (20)                     |
| E       | (105)                    |

10. ABC Limited has 5 segments namely A, B, C, D and E. The profit/loss of each segment for the year ended March 31<sup>st</sup>, 20X2 is as follows:

| Segment | Profit/(Loss)<br>(` in crore) |
|---------|-------------------------------|
| A       | 780                           |
| B       | 1,500                         |
| C       | (2,300)                       |
| D       | (4,500)                       |
| E       | 6,000                         |
| Total   | 1,480                         |

Identify the Reportable segments.

11. Heavy Goods Ltd. has 6 segments namely L-Q (below).

The total revenues (internal and external), profits or losses and assets are set out below:

(In `)

| Segment | Inter Segment Sales | External Sales | Profit / loss | Total assets |
|---------|---------------------|----------------|---------------|--------------|
| L       | 4,200               | 12,300         | 3,000         | 37,500       |
| M       | 3,500               | 7,750          | 1,500         | 23,250       |
| N       | 1,000               | 3,500          | (1,500)       | 15,750       |
| O       | 0                   | 5,250          | (750)         | 10,500       |
| P       | 500                 | 5,500          | 900           | 10,500       |
| Q       | 1,200               | 1,050          | 600           | 5,250        |
|         | 10,400              | 35,350         | 3,750         | 1,02,750     |

Heavy Goods Ltd. needs to determine how many reportable segments it has.

You are required to advise Heavy Goods Ltd. as per the criteria defined in AS 17.

12. Calculate the segment results of a manufacturing organization from the following information:

| Segments                                             | A        | B        | C        | Total    |
|------------------------------------------------------|----------|----------|----------|----------|
| Directly attributed revenue                          | 5,00,000 | 3,00,000 | 1,00,000 | 9,00,000 |
| Enterprise revenue<br>(allocated in 5 : 4 : 2 basis) |          |          |          | 1,10,000 |
| Revenue from transactions with other segments        |          |          |          |          |
| Transaction from B                                   | 1,00,000 |          | 50,000   | 1,50,000 |
| Transaction from C                                   | 10,000   | 50,000   |          | 60,000   |
| Transaction from A                                   |          | 25,000   | 1,00,000 | 1,25,000 |

|                                                        |          |          |        |          |
|--------------------------------------------------------|----------|----------|--------|----------|
| Operating expenses                                     | 3,00,000 | 1,50,000 | 75,000 | 5,25,000 |
| Enterprise expenses<br>(allocated in 5 :4 :2<br>basis) |          |          |        | 77,000   |
| Expenses on transactions with<br>other segments        |          |          |        |          |
| Transaction from B                                     | 75,000   |          | 30,000 |          |
| Transaction from C                                     | 6,000    | 40,000   |        |          |
| Transaction from A                                     |          | 18,000   | 82,000 |          |

13. The Senior Accountant of AMF Ltd. gives the following data regarding its five segments:

(` in lakhs)

| Particulars     | P     | Q   | R   | S    | T   | Total |
|-----------------|-------|-----|-----|------|-----|-------|
|                 | (`)   | (`) | (`) | (`)  | (`) | (`)   |
| Segment Assets  | 80    | 30  | 20  | 20   | 10  | 160   |
| Segment Results | (190) | 10  | 10  | (10) | 30  | (150) |
| Segment Revenue | 620   | 80  | 60  | 80   | 60  | 900   |

The Senior Accountant is of the opinion that segment "P" alone should be reported. Is he justified in his view? Examine his opinion in the light of provision of AS-17 'Segment Reporting'.

## ANSWERS/ HINTS

### Answers to the Multiple Choice Questions

|    |     |    |     |    |     |    |     |    |     |
|----|-----|----|-----|----|-----|----|-----|----|-----|
| 1. | (a) | 2. | (c) | 3. | (a) | 4. | (a) | 5. | (c) |
|----|-----|----|-----|----|-----|----|-----|----|-----|

### Answers to the Scenario based Question

6. According to AS 17 "Segment Reporting", segment assets do not include income tax assets.

Therefore, the revised total assets are 12.3 crores [ $\text{₹ } 15 - (\text{₹ } 1 + 0.9 + 0.8)$ ].

*Details of Segment wise assets:*

Segment P holds total assets of ₹ 3 crores (₹ 4 crores - ₹ 1 crores); Segment Q holds ₹ 5.1 crores (₹ 6 crores - ₹ 0.9 crores);

Segment R holds ₹ 4.2 crores (₹ 5 crores - ₹ 0.8 crores).

Thus, all the three segments hold more than 10% of the total assets, all segments are reportable segments.

Hence, the contention of the Accountant that all three segments are reportable segments is correct.

7. AS 17 explains that, “a single geographical segment does not include operations in economic environments with significantly differing risks and returns. A geographical segment may be a single country, a group of two or more countries, or a region within a country”.

Accordingly, to identify geographical segments, Company A needs to evaluate whether the segments reflected in the management information system function in environments that are subject to significantly differing risks and returns irrespective of the fact whether they are within the same country.

The Standard recognizes that, “Determining the composition of a business or geographical segment involves a certain amount of judgement...”. Accordingly, while the management information system of the Company provides segment information for rural and urban geographical segments for the purpose of internal reporting, judgement is required to determine whether these segments are subject to significantly differing risks and returns based on the definition of geographical segment. In making such a judgement, aspect like different pricing and other policies, e.g., credit policies, deployment of resources between different regions etc., may be considered for the purpose identifying ‘urban and ‘rural’ as separate geographical segment.

Company A, in making judgment for identifying geographical segments, should also consider the relevance, reliability and comparability over time of segment information that will be reported. The Standard, explains that, “In making that judgement, enterprise management takes into account the

objective of reporting financial information by segment as set forth in the standard and the qualitative characteristics of financial statements. The qualitative characteristics include the relevance, reliability and comparability over time of financial information that is reported about the different groups of products and services of an enterprise and about its operations in particular geographical areas, and the usefulness of that information for assessing the risks and returns of the enterprise.”

- 8.** As per AS 17, if primary format of an enterprise for reporting segment information is business segments, it should also report segment revenue from external customers by geographical area based on the geographical location of its customers, for each geographical segment whose revenue from sales to external customers is 10 per cent or more of enterprise revenue.

Therefore, for the purposes of disclosing secondary segment information, PK Ltd. is not required to disclose segment revenue from export sales to UK, since that segment does not meet the 10 per cent or more of enterprise revenue threshold. However, other secondary segment information as per AS 17 should be disclosed in respect of this segment if the thresholds prescribed in the AS 17 are met.

- 9.** As per AS 17 ‘Segment Reporting’, a business segment or geographical segment should be identified as a reportable segment if:

Its segment results whether profit or loss is 10% or more of:

- ♦ The combined result of all segments in profit; i.e. ₹ 250 Lakhs or
- ♦ The combined result of all segments in loss; i.e. ₹ 300

Lakhs Whichever is greater in absolute amount i.e. ₹ 300

| Operating Segment | Absolute amount of Profit or Loss (₹ In lakhs) | Reportable Segment Yes or No |
|-------------------|------------------------------------------------|------------------------------|
| A                 | 225                                            | Yes                          |
| B                 | 25                                             | No                           |
| C                 | 175                                            | Yes                          |

On the basis of the profitability test (result criteria), segments A, C and E are reportable segments (since their results in absolute amount is 10% or more of ₹ 300 lakhs i.e. 30 lakhs).

**10.** In compliance with AS 17, the segment profit/loss of respective segment will be compared with the greater of the following:

- (i) All segments in profit, i.e., A, B and E - Total profit ₹ 8,280 crores.
- (ii) All segments in loss, i.e., C and D - Total loss ₹ 6,800 crores.

Greater of the above - ₹ 8,280 crores.

Based on the above, reportable segments will be determined as follows:

| Segment      | Profit/(Loss) | Absolute Profit/Loss as a % of 8,280 | Reportable Segment |
|--------------|---------------|--------------------------------------|--------------------|
| A            | 780           | 9%                                   | No                 |
| B            | 1,500         | 18%                                  | Yes                |
| C            | (2,300)       | 28%                                  | Yes                |
| D            | (4,500)       | 54%                                  | Yes                |
| E            | 6,000         | 72%                                  | Yes                |
| <b>Total</b> | <b>1,480</b>  |                                      |                    |

**11.** Quantitative Threshold Test:

**Revenue Test:**

Combined total sales of all the segment = ₹ 10,400 + ₹ 35,350 = ₹ 45,750. 10% thresholds =  $45,750 \times 10\% = 4,575$ .

**Profitability Test:**

In the given situation, combined reported profit = ₹ 6,000 and combined reported loss (₹ 2,250). Hence, for 10% thresholds ₹ 6,000 will be considered.

10% thresholds =  $₹ 6,000 \times 10\% = ₹ 600$

**Asset Test:**

Combined total assets of all the segment = ` 1,02,750  
 10% thresholds = ` 1,02,750 x 10% = 10,275

**Accordingly, quantitative thresholds are calculated below:**

| Segments                          | L      | M      | N      | O      | P      | Q     | Reportable segments |
|-----------------------------------|--------|--------|--------|--------|--------|-------|---------------------|
| % segment sales to total sales    | 36.66% | 24.59% | 9.84%  | 11.48% | 13.11% | 4.92% | L, M,O,P            |
| % segment profit to total profits | 50%    | 25%    | 25%    | 12.5%  | 15%    | 10%   | L,M,N,O,P, Q        |
| % segment assets to total assets  | 36.50% | 22.63% | 15.33% | 10.22% | 10.22% | 5.11% | L,M,N,O,P           |

**Conclusion:**

Segments L, M, O and P clearly satisfy the revenue and assets tests and they are separate reportable segments.

Segment N does not satisfy the revenue test, but it does satisfy the asset test and it is a reportable segment.

Segment Q does not satisfy the revenue or the assets test but it does satisfy the profits test. Therefore, Segment Q is also a reportable segment.

Hence all segments i.e. L, M, N, O, P and Q are reportable segments.

**12.**

**Computation of segment result:**

| Segments                    | A        | B        | C        | Total    |
|-----------------------------|----------|----------|----------|----------|
| Directly attributed revenue | 5,00,000 | 3,00,000 | 1,00,000 | 9,00,000 |

|                                                       |        |        |        |          |
|-------------------------------------------------------|--------|--------|--------|----------|
| Enterprise revenue<br>(allocated in 5 :4 :2<br>basis) | 50,000 | 40,000 | 20,000 | 1,10,000 |
|-------------------------------------------------------|--------|--------|--------|----------|

|                                                      |                 |                 |                 |                  |
|------------------------------------------------------|-----------------|-----------------|-----------------|------------------|
| <b>Revenue from transactions with other segments</b> |                 |                 |                 |                  |
| Transaction from B                                   | 1,00,000        |                 | 50,000          | 1,50,000         |
| Transaction from C                                   | 10,000          | 50,000          |                 | 60,000           |
| Transaction from A                                   |                 | 25,000          | 1,00,000        | 1,25,000         |
| <b>Total segment revenue (1)</b>                     | <b>6,60,000</b> | <b>4,15,000</b> | <b>2,70,000</b> | <b>13,45,000</b> |
|                                                      | <b>0</b>        | <b>0</b>        | <b>0</b>        | <b>0</b>         |
| Operating expenses                                   | 3,00,000        | 1,50,000        | 75,000          | 5,25,000         |
|                                                      |                 | 0               |                 |                  |
| Enterprise expenses (allocated in 5 :4 :2 basis)     | 35,000          | 28,000          | 14,000          | 77,000           |
| Expenses on transactions with other segments         |                 |                 |                 |                  |
| Transaction from B                                   | 75,000          |                 | 30,000          | 1,05,000         |
| Transaction from C                                   | 6,000           | 40,000          |                 | 46,000           |
| Transaction from A                                   |                 | 18,000          | 82,000          | 1,00,000         |
| <b>Total segment expenses (2)</b>                    | <b>4,16,000</b> | <b>2,36,000</b> | <b>2,01,000</b> | <b>8,53,000</b>  |
|                                                      | <b>0</b>        | <b>0</b>        | <b>0</b>        |                  |
| <b>Segment result (1-2)</b>                          | <b>2,44,000</b> | <b>1,79,000</b> | <b>69,000</b>   | <b>4,92,000</b>  |
|                                                      | <b>0</b>        | <b>0</b>        |                 |                  |

**13.** As per AS 17 'Segment Reporting', a business segment or geographical segment should be identified as a reportable segment if:

- (i) Its revenue from sales to external customers and from other transactions with other segments is 10% or more of the total revenue- external and internal of all segments; or
- (ii) Its segment result whether profit or loss is 10% or more of:
  - (1) The combined result of all segments in profit; or
  - (2) The combined result of all segments in loss, whichever is greater in absolute amount; or
- (iii) Its segment assets are 10% or more of the total assets of all segments. Accordingly,
  - (a) On the basis of revenue from sales criteria, segment P is a reportable segment.

- (b) On the basis of the result criteria, segments P & T are reportable segments (since their results in absolute amount is 10% or more of ₹ 200 Lakhs).
- (c) On the basis of asset criteria, all segments except T are reportable segments.

Since all the segments are covered in at least one of the above criteria, all segments have to be reported upon in accordance with AS 17. Hence, the opinion of chief accountant that only segment 'P' is reportable is wrong.